

JOWLER CREEK BRIDGE
Spanning Jowler Creek at Interurban Road
Camden Point vicinity
Platte County
Missouri

HAER No. MO-92

HAER
MO-92

PHOTOGRAPHS

WRITTEN HISTORICAL AND DESCRIPTIVE DATA

HISTORIC AMERICAN ENGINEERING RECORD
National Park Service
U.S. Department of the Interior
1849 C St. NW
Washington, DC 20240

HISTORIC AMERICAN ENGINEERING RECORD
JOWLER CREEK BRIDGE

HAER No. MO-92

Location: Interurban Road over Jowler Creek
1.7 miles south of State Route E
Camden Point vicinity
Platte County, Missouri

UTM: 15.349180.43650620
Quad: Tracy, Missouri, 1:24,000

Date of Construction: 1912-13

Engineer: Robert P. Woods

Architect: Standard design by Daniel Luten

Present Owner: Platte County, Missouri

Present Use: Vehicular Bridge

Significance: The Jowler Creek Bridge is one of the two-arch concrete bridges of the Luten design built for the Kansas City, Clay County and St. Joseph Railway, an electric interurban railway that operated from 1913-33. The railway was the first of its kind to offer service between the two largest cities in northwest Missouri, Kansas City and St. Joseph. In addition, it played a significant role in the development of the suburban communities in the region.

Project Information: An engineering recommendation in 1993 advised the replacement of the bridge with a structure that will provide a wider roadway. To mitigate the adverse effect, the Missouri State Historic Preservation Office stipulated documentation of the bridge.

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Summary of Bridge and Setting

This bridge carries vehicular traffic over Jowler Creek on the Interurban Road, 1.7 miles south of Camden Point in Platte County. Built in 1912-13 for use as an electric railway bridge, it is a two-span filled concrete arch structure of the Luten design. Each span is 47.5' in length; the overall bridge length is 95'. The 18.1' wide structure has a 16.4' roadway width. The bridge is 23' in height over the wooded creek bottom, which is surrounded by farmland. A farmstead with a nineteenth century house is located on the northeast side of the bridge and creek.

The Interurban Road follows 16.5 miles of the original 45.88-mile route of the St. Joseph line of the former Kansas City, Clay County and St. Joseph (KCCC&SJ) Railway. The road extends from the north edge of the town of Ferrelview on the south to the south edge of the town of Dearborn on the north.

The 1913-33 electric interurban railway was an important transportation system that operated in northwest Missouri because it was the first of its kind to provide fast, high-speed passenger and freight service between Kansas City and the areas north of the Missouri River. It also was one of the state's longest electric interurban railways. To successfully compete against existing steam railroads, the railway's owners financed the construction of the straightest, shortest possible route and established several stations along the route. Carrying out the plan required cutting across the rugged, hilly terrain of a rural region dotted with small towns, farms, and woodland. The company operated two lines: one to St. Joseph; the other to Excelsior Springs. Over those lines, the KCCC&SJ provided an efficient travel connection between the two largest cities of the region, Kansas City and St. Joseph; at the same time, it served as an important component of the development of growing suburban communities in Platte, Buchanan and Clay Counties.

The railway ceased operations in 1933 due to financial difficulties and was sold at auction in 1934. A salvage company purchased the metals and, in Platte County, had completed its salvage work before December 1935. The concrete bridges were left in place after the tracks were removed; many of them were converted, along with the old railway road bed, into a farm-to-market road between 1935-40.

History of the Kansas City, Clay County and St. Joseph Railway

The KCCC&SJ Railway, constructed between 1911-13 by the Wyandotte Construction Company, Kansas City, was the first high-speed electric interurban service to connect Kansas City with towns and cities located north of the Missouri River in northwest Missouri. The idea for such a line had been considered for some time, but no plan had been carried out previously because the cost of construction was considered too great to make a railway a profitable venture. The main obstacle was the need to construct a new bridge over the Missouri River. Although the steam railroads that operated in the region had bridges over the river, none was willing to grant trackage rights to a new, competing electric system. However, the construction of the Armour-Swift-Burlington (ASB) Bridge, completed in 1909, changed the investment climate. The builders of the ASB Bridge also had purchased a large tract of land on the river's north side with plans of development. They recognized interurban service would complement their development plans, thus, granted trackage rights on the new bridge. Eastern capitalists (primarily banking houses) then were persuaded to finance the formation of the railway company and to build the interurban. The electric rail service encouraged population growth in existing communities along its route. Two new towns, Glenaire and Northmoor, were established.¹

The company secured franchises from the counties and cities it would serve with the Platte County Court granting a 200-year franchise on September 11, 1911. Under terms of the franchise, the company agreed to purchase needed right-of-way from landowners (between 66' and 100' wide) and to build all necessary bridges.²

The railway's route originated in downtown Kansas City and, after crossing the Missouri River, divided into two lines: one to Excelsior Springs, the site of a popular mineral water spa of the era located in Clay County; the second ran to St. Joseph, the second largest city in the region. The Excelsior Springs line, 26.45 miles in length, was completed first, with the first car running on January 21, 1913; the 45.88-mile St. Joseph line was opened five months later, on May 5.³ The total mileage, 72.33, comprised nearly half the total number of miles built in Missouri during the height of the interurban construction era (1893-1915).⁴

The company purchased its electric power from the Metropolitan Street Railway Company, Kansas City, which transmitted the current to the KCCC&SJ transformer station on the north side of the river. From there, the power was sent to three substations on the two lines and from the substations to the overhead lines that powered the cars.⁵

Passenger stations and shelter stations were built along the line at several locations. A 1927 schedule listed forty-one stops between Kansas City and St. Joseph; most were at small shelters. Small stations, operated by local individuals, were located in Platte County at Ferrelview, Keerville, Camden Point and Dearborn. The train cars also stopped for anyone who flagged them down along the route.⁶

The KCCC&SJ ran on regular one-hour schedules from early morning until midnight, the last train carried passengers back along the St. Joseph route from late movies in downtown Kansas City. The service attracted a variety of customers, giving them speedy access to and from the region's major metropolitan area. For urban workers who lived in the developing suburbs, it provided transportation to jobs in an era before automobiles became affordable to the working class. Farmers used the train to transport produce to the Kansas City and St. Joseph markets. Students rode it to schools. It also provided a means of transportation to shopping and entertainment centers of the area.⁷

The interurban enjoyed moderate success during its early years of operation, although it never achieved the kind of financial return that was expected of it by investors. It was the coming of the automobile, the construction of hard surface roads and the financial effects of a number of lawsuits filed against the company, however, that caused the company's net earnings to decline in the mid-1920s.⁸ By 1930, with the onset of the Depression, the company's position worsened and, in September that year, it defaulted on bonds. Two months later, in November, a federal judge placed the railway into a receivership. Soon after, the receiver filed a petition with the Missouri Public Service Commission (PSC), asking for permission to discontinue service. The PSC granted that permission on February 12, 1933. A judge in the U.S. District Court for the Western Division of the Western District then set the date the railway's operation would end: March 10, 1933.⁹ The last cars on both lines ran that day.¹⁰

Several efforts were made to revive the railway, but all failed. One of the most successful attempts was headed by Ben M. Achtenberg of Kansas City, an attorney who saw the need for keeping the fast rail connection in operation. While many of those who used the KCCC&SJ expressed great support for its continuation, there was very little financial support given except by Achtenberg. He succeeded

in obtaining a certificate of convenience and necessity from the PSC, granted on July 7, 1934 after hearings were conducted in April that year in Jefferson City.¹¹ However, Achtenberg was unable to raise sufficient capital by the time the railway was sold at a court-ordered auction that began on August 21, 1934. The Hymen-Michaels Salvage Company of Chicago offered the winning bid, \$112,525 for the railway's equipment, excluding stations, electric substation structures and rolling stock.¹² Within days, salvage crews began the removal of the metals from the railway (primarily iron from tracks and bridges and copper wire from the railroad's electric system.). The St. Joseph line was dismantled first, but the entire salvage project was completed by the end of 1935.¹³

The Bridge

The Jowler Creek Bridge is one of the KCCC&SJ Railways' 108 Luten design concrete bridges, built by the Wyandotte Construction Company from standard engineering plans originally drawn by Daniel Luten.¹⁴ The project's chief engineer, Robert P. Woods, had built Luten arch bridges for similar interurban systems in other parts of the United States, including in Indiana. He used the same plans to build the KCCC&SJ. In addition to the Luten design concrete arch bridges, six steel bridges were constructed for the line.¹⁵

Located on the project's alignment and profile maps at the 22-mile point, the two-span bridge is among the reinforced concrete arch structures required for crossing larger waterways.¹⁶ All the railway's bridges were designed to carry concentrated loads of 100 tons. Both the structure and the deck are of concrete. The cost of a Luten arch was determined by the formula: $I(S+2)(3F+W)=10$. S=span in feet; F=profile fill; W=Subgrade width in feet; I=Unity for these prices: cement, \$1.25 per yard; sand, \$1 per yard; stone, \$1 per yard; lumber, \$2 per feet; steel, \$1.75 per hundredweight; labor, \$2 per day; excavation, \$.50 per yard and superintendent, \$.50 per hour.¹⁷

No historic photographs have been found that identify the Jowler Creek Bridge. However, there are photographs identified as KCCC&SJ bridges (but not named as specific sites), that appear to be very similar to the Jowler Creek structure. Photographs published in a book about Daniel Luten arches include four built for the KCCC&SJ. None is the Jowler Creek Bridge, but one, located at Liberty on the Excelsior Springs line, is quite similar.¹⁸ The photos were published as examples of the 108 concrete bridges.

No county road records were found that contain information about specific work projects performed on the bridge since the time it was converted from a railway to a vehicular structure. The bridge is not posted. However, signage that denotes it as a single-lane bridge is set at both north and south approaches. The roadway and bridge deck have asphalt surfaces at the present time.

The Wyandotte Construction Company

The Wyandotte Construction Company was formed in 1910 for the sole purpose of constructing the KCCC&SJ Railway. Its principals were George Townsend, Walter Kirkpatrick and Ward S. Arnold. The company is not known to have constructed any other project and ceased operations when its assets were dissolved in June 1913.¹⁹

The company controlled and oversaw the entire interurban project, subcontracting some of the work.²⁰ Some of the workers along the St. Joseph line were farmers who needed to subsidize their income because of a crop failure in 1912. Local workers also were hired in numbers to help with the grading and bridge-building phases of the project, in part, because of transportation curtailments of the day. Horses and mules were used for the heaviest work. However, when it was time to lay ties and rails, crews from Kansas City rode hand-pump cars to the site, carrying with them the materials needed for the day's work. In all, an estimated 1,500 men worked on the construction project.²¹

Daniel Luten

Daniel B. Luten was a nationally-recognized civil engineer of the late nineteenth and early twentieth centuries. Granted a number of patents for his bridge designs, Luten particularly was known for his work with reinforced concrete bridges and arches. He is credited with overseeing the construction of at least 12,000 bridges across the U.S. during his career. He also taught, published professional papers and articles and presented illustrated lectures to professional engineering groups and university engineering departments.²²

Robert P. Woods

Robert P. Woods was the one individual who is most closely associated with the KCCC&SJ Railway. As chief engineer, he was responsible for the construction of the railway. Two decades later, he was the individual who oversaw its operations cease, auctioned and dismantled. Before he came to Kansas City, Woods had overseen the construction of similar interurban systems in other parts of the country and had worked on various other types of engineering projects in the U.S. as well as in Mexico. He was hired by the KCCC&SJ Railway Company as the project's chief engineer in 1910. In 1920, he was named its general manager. Woods was named the railway company's receiver by a federal judge in 1930. After the KCCC&SJ was auctioned, Woods became Street Railway Commissioner of Kansas City, Missouri.²³

The Interurban Road

After the railway was dismantled in Platte County, the ties, at least one iron bridge and the concrete bridges and culverts were left behind. The Platte County Court purchased the iron bridge from the salvage company in March 1935.²⁴ No record has been found of county purchase or acceptance of the Jowler Creek Bridge or of any of the concrete bridges and culverts on the road. County court records do show however, that, between 1935-41, petitions were filed by landowners in the area who asked the county to convert the railway road into a farm-to-market route which is today locally known as the Interurban Road. By early 1935, part of the road project had begun.²⁵ The bridges along the 16.5 mile route were converted in that project, but exact dates of the various phases of construction, most specifically for the Jowler Creek Bridge and its immediate area, have not been determined.

The Project

Housing development has increased over the years along the Interurban Road which now carries more traffic than in previous years. The Jowler Creek Bridge has been designated as a single-lane bridge in recent years.

In an update to the county commission in 1993, the county engineer recommended the bridge be replaced with a modern structure wide enough for two lanes of traffic.²⁶

Notes

1. "The Kansas City, Clay County and St. Joseph Railway," *Electric Railway Journal*, 42, (August 9, 1913): 212.; H. Roger Grant, "The Excelsior Springs Route: Life and Death of a Missouri Interurban," *Missouri Historical Review*, 65, No. 1 (October 1970): 38-39.
2. Platte County, Missouri Road Record, Book D, Sept. 9, 1911.253.
3. "KCCC&SJ Railway," 212; *St. Joseph (Missouri) News-Press*, May 5, 1913.
4. Grant, "The Excelsior Springs Route," 38.
5. "KCCC&SJ Railway," 216.
6. Betty Soper, "Old Interurban in Platte County," *Platte County Historical Society and Genealogical Society Quarterly*, 34 (Winter 1981): 11.
7. Ibid, 10-12; Grant, "The Excelsior Springs Route," 41-42.
8. Missouri Public Service Commission, *Public Service Commission Reports*, Jefferson City, 1934.124.
9. W.S. Tuley, Treasurer for the (KCCC&SJ) Receiver, "History of Discontinuance of Service" prepared for Joseph B. Eastman, Federal Coordinator of Transportation, Washington, D.C. August 23, 1933. This compilation, along with numerous copies of PSC and court orders pertaining to the railway are found in Box 9, KC302, Terry Cassidy Collection, Western Historical Manuscript Collection-Kansas City, Missouri.
10. *St. Joseph (Missouri) News-Press*, March 10, 1933. Platte City (Missouri) *Landmark*, March 10, 1933.
11. Missouri Public Service Commission. Transcripts of public hearing on Case 8710, Application for Certificate of Necessity and Convenience, filed by B.M. Achtenberg, Kansas City, April 17-20, 1934; Commission Order on Case 8710, July 7, 1934.
12. *Kansas City Times*, August 23, 1934; Special Master's Supplemental Report on Completion of Sale, October 8, 1934, Cassidy Collection, WHMC-KC.
13. Shelby Singleton, former employee of Hymen-Michaels Co., who worked on the KCCC&SJ salvage project, interview by Mary Ann Lowary at Dearborn, Missouri (October 1995); *Kansas City Times*, April 25, 1935.
14. "KCCC&SJ Railway," 214; Robert P. Woods, "Character of the Kansas City, Clay County and St. Joseph Railway, Missouri," Short Line, Inc. Collection (MSLC), Kansas City, Missouri, Undated.

15. David Short, Missouri Short Line, Inc., interview by Mary Ann Lowary at Kansas City, Missouri (September 1995).
16. Wyandotte Construction Co. *Alignment and Profile Map, St. Joseph Division of Kansas City, Clay County and St. Joseph Railway*, June 1913, Sheet 13, MSLC.
17. "KCCC&SJ Railway," 214-215.
18. There are a number of photographs of both the construction of and features of the KCCC&SJ Railway, including photographs of its concrete arch bridges. Two views of a double-arch bridge that is similar to the Jowler Creek structure (in a similar setting) are among seven bridge views included in one unnamed negative (N-118-412), Cassidy Collection, WHMC-KC; *Daniel Luten Arches*, [title/copyright page missing], Robert P. Woods Papers, MSLC.; Private collection of Robert Creek, Richmond, Missouri, contains several photographs of the construction of the railway taken by George Creek, but none are identified as the Jowler Creek site. No photographs of the bridge were found in the Kansas City Museum, the St. Joseph Museum or at the Platte County Museum in Platte City, Missouri.
19. David Short interview.
20. "KCCC&SJ Railway," 221.
21. "Old Interurban in Platte County." 9.
22. *Daniel Luten Arches*. 5.
23. David Short interview.
24. Platte County Missouri Road Record. Book G, March 18, 1935. 295; Platte City (Missouri) *Landmark*, March 29, 1935; *Kansas City Times*, April 25, 1995.
25. *Kansas City Times*, April 25, 1935.; More than 10 contacts were made with older residents of the area and descendants of former residents. All say the road was converted during the period of 1935 to early 1940s, but cannot recall the year of the work done on the Jowler Creek site. Various entries recorded in the Platte County Court Road Record during those years are of petitions of residents who requested the old railway be converted. The spread of dates suggest it was a project completed in phases during the year.
26. Platte County Commission, *Minutes*, Oct. 12, 1993.2.

Bibliography

Cassidy, Terry Collection, Box 9, KC302, Western Historical Manuscript Collection-Kansas City, Missouri
Various papers, documents, copies of documents, photographs, notes.

Grant, H. Roger, "The Excelsior Springs Route: Life and Death of a Missouri Interurban, *Missouri Historical Review*, 65, no. 1 (October 1970): 37-50.

"The Kansas City, Clay County and St. Joseph Railway", *Electric Railway Journal*, 42, (August 9, 1913): 212-221.

Kansas City Times, August 12, 1934; April 25, 1934; April 25, 1935.

Luten, Daniel B. Arches, [title/copyright page missing]. 2.19-20.35.

Landmark, Platte City (Missouri), March 29, 1935.

Missouri Public Service Commission, Transcripts of public hearing on Case 8710, Jefferson City, (April 1934).

Missouri Public Service Commission, Commission order on Case 8710, Jefferson City, (July 1934).

Platte County Missouri Road Record, Book D. Sept. 9, 1911. 253.

Platte County Missouri Road Record, Book G., March 18, 1935. 295.

Platte County Missouri Commissioners Minutes, October 10, 1993.

St. Joseph (Missouri) News-Press, March 10, 1933.

Short, David W. Missouri short Line, Inc., interview September 29, 1995, Kansas City, Missouri

Singleton, Shelby. Interview by Mary Ann Lowary. Dearborn, Missouri (October 7, 1995).

Soper, Betty, "Old Interurban in Platte County, *Platte County Historical Society and Genealogical Society Quarterly*, 34 (Winter 1981): 9-15.

Tuley, W.S. "History of Discontinuance of Service." Box 9,. KC302, Terry Cassidy Collection, Western Historical Manuscript Collection-Kansas City, Missouri

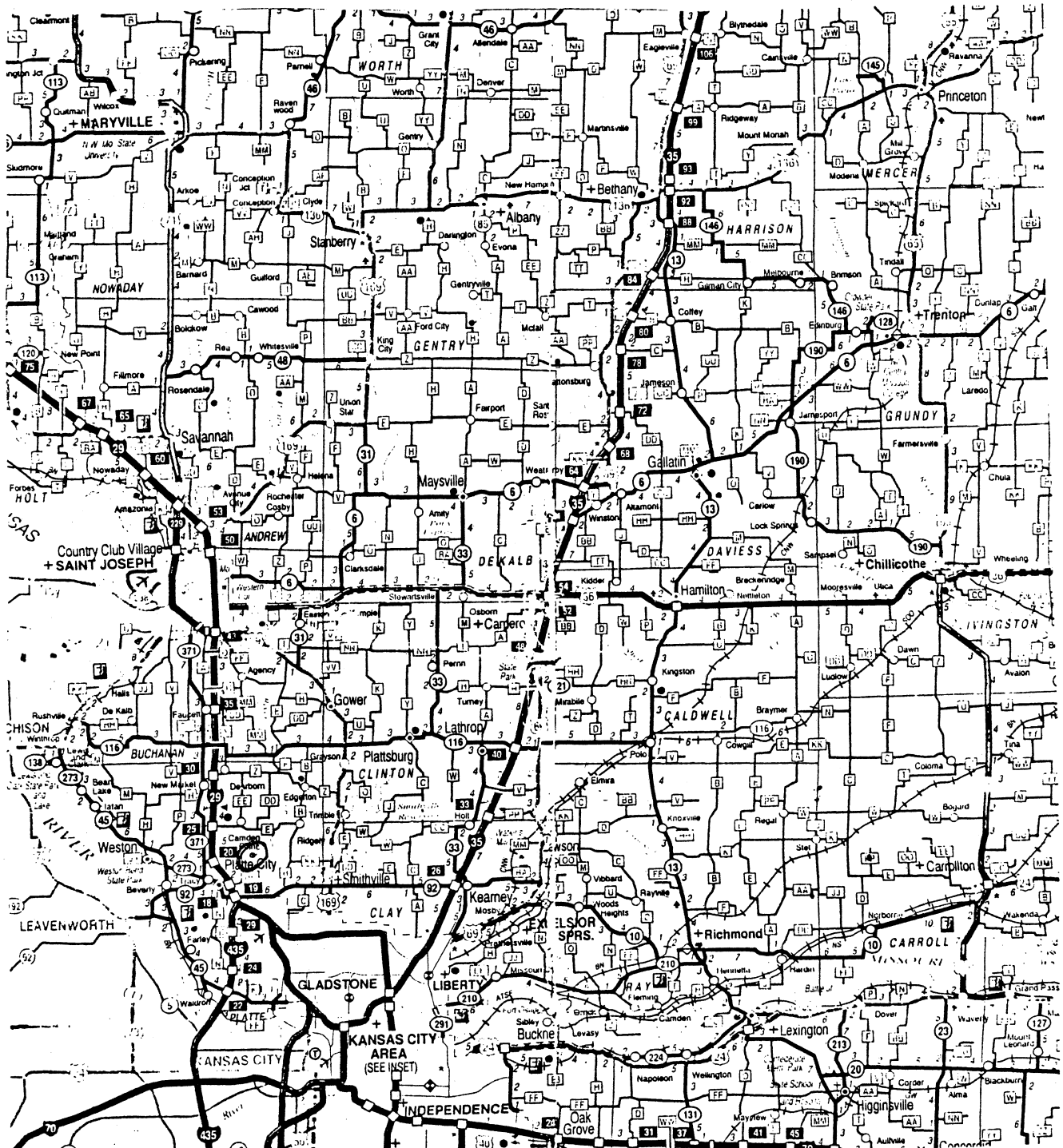
Woods, Robert P. "Character of Construction of the Kansas City, Clay County and St. Joseph Railway," Missouri Short Line, Inc. Collection, Kansas City, Missouri

Woods, Robert P. Papers, collection of Short Line, Inc. Kansas City, Missouri various papers, documents, clippings reviewed in uncatalogued collection.

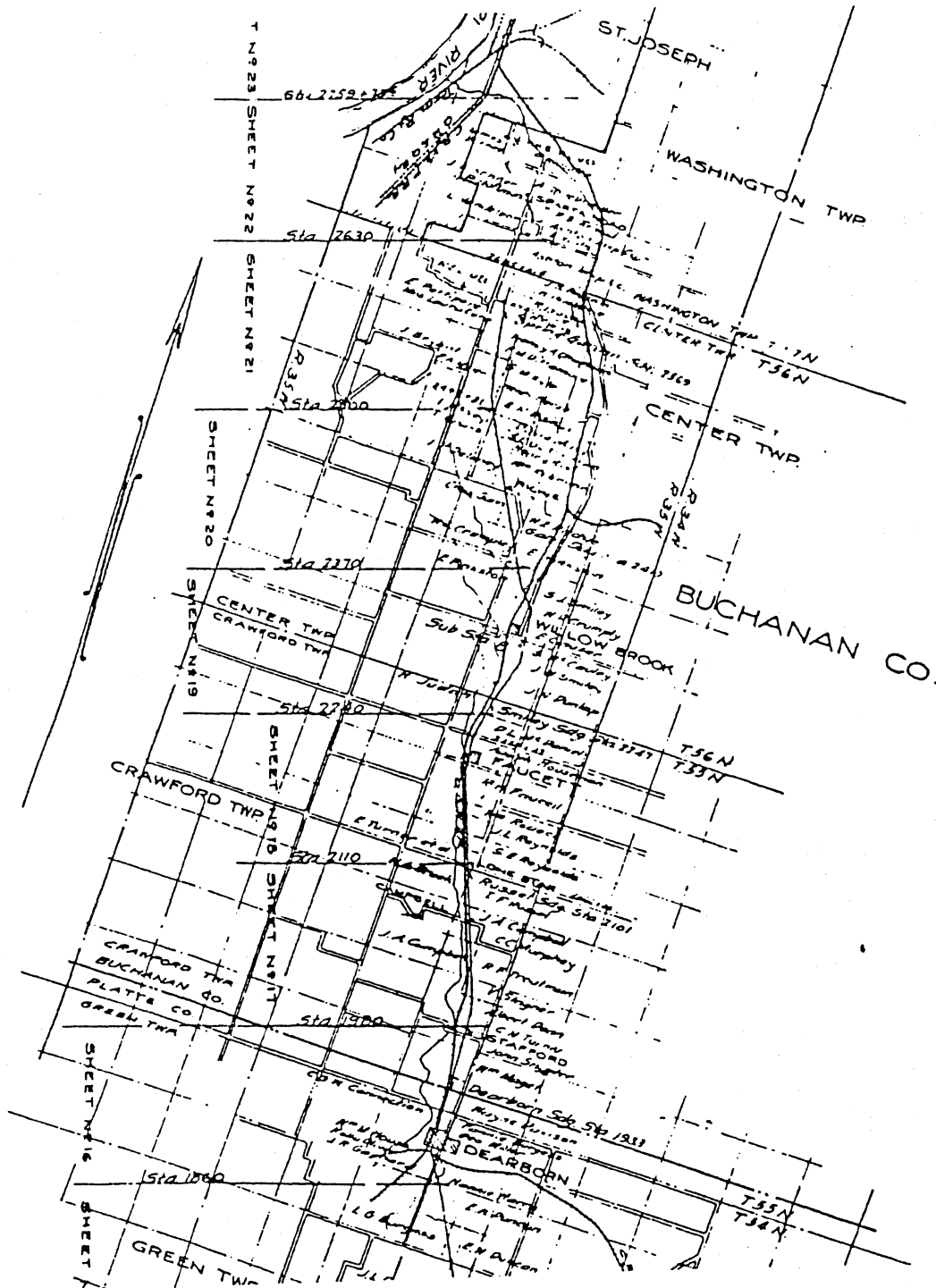
Bibliography, continued

Wyandotte Construction Co., Kansas City, Missouri *Alignment and Profile Map, St. Joseph Division of Kansas City, Clay County and St. Joseph Railway*, (June 1913), MSLC.

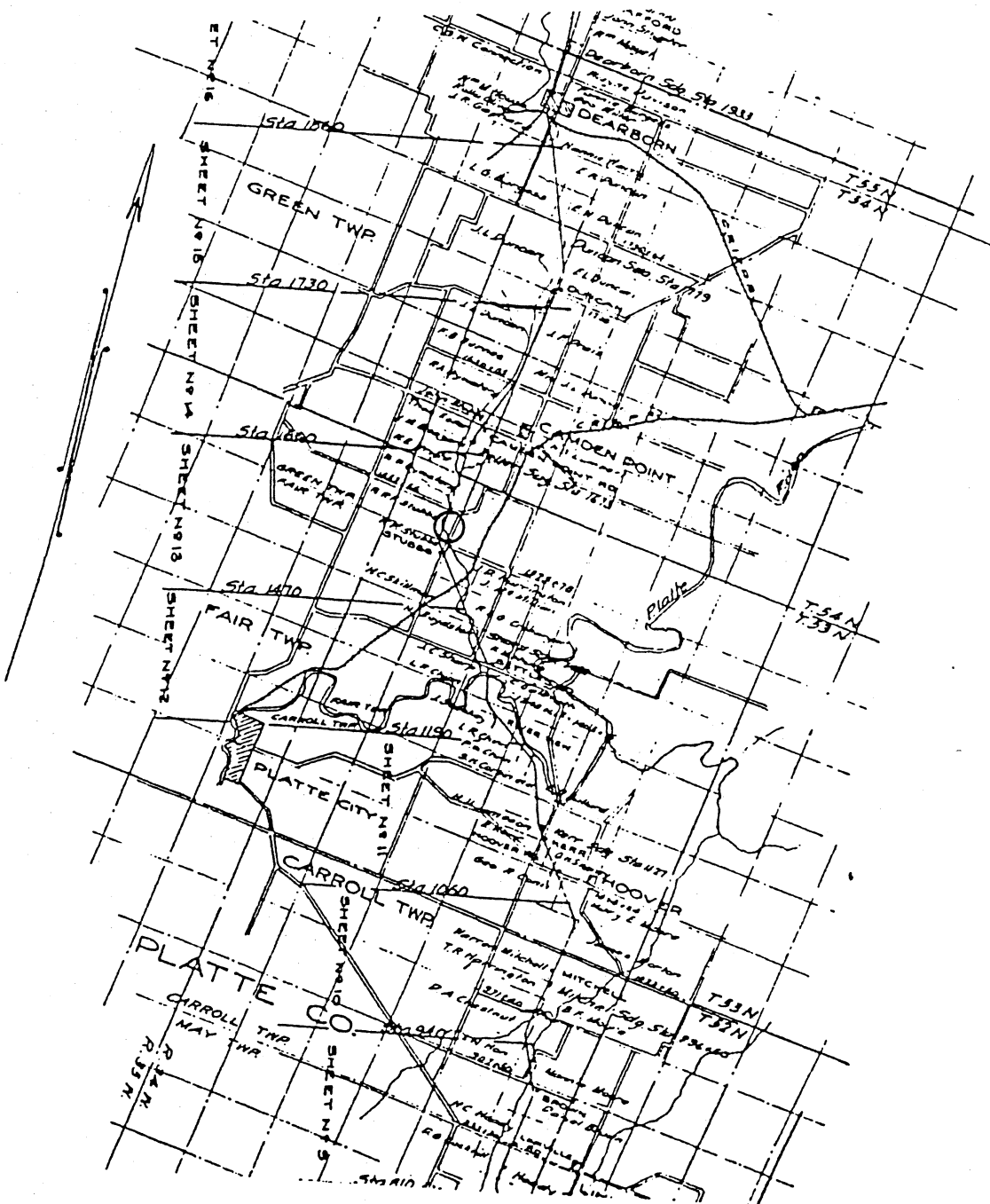
Present day map showing location of Jowler Creek Bridge, Camden Point vicinity, Platte County, Missouri.



Wyandotte Construction Company Index to Profile & Alignment Maps, June 1913, showing the location of the Jowler Creek Bridge on the St. Joseph line of the KCCC&SJ Railway. Part 1 of 3 (Parts 2 and 3 on two following pages)



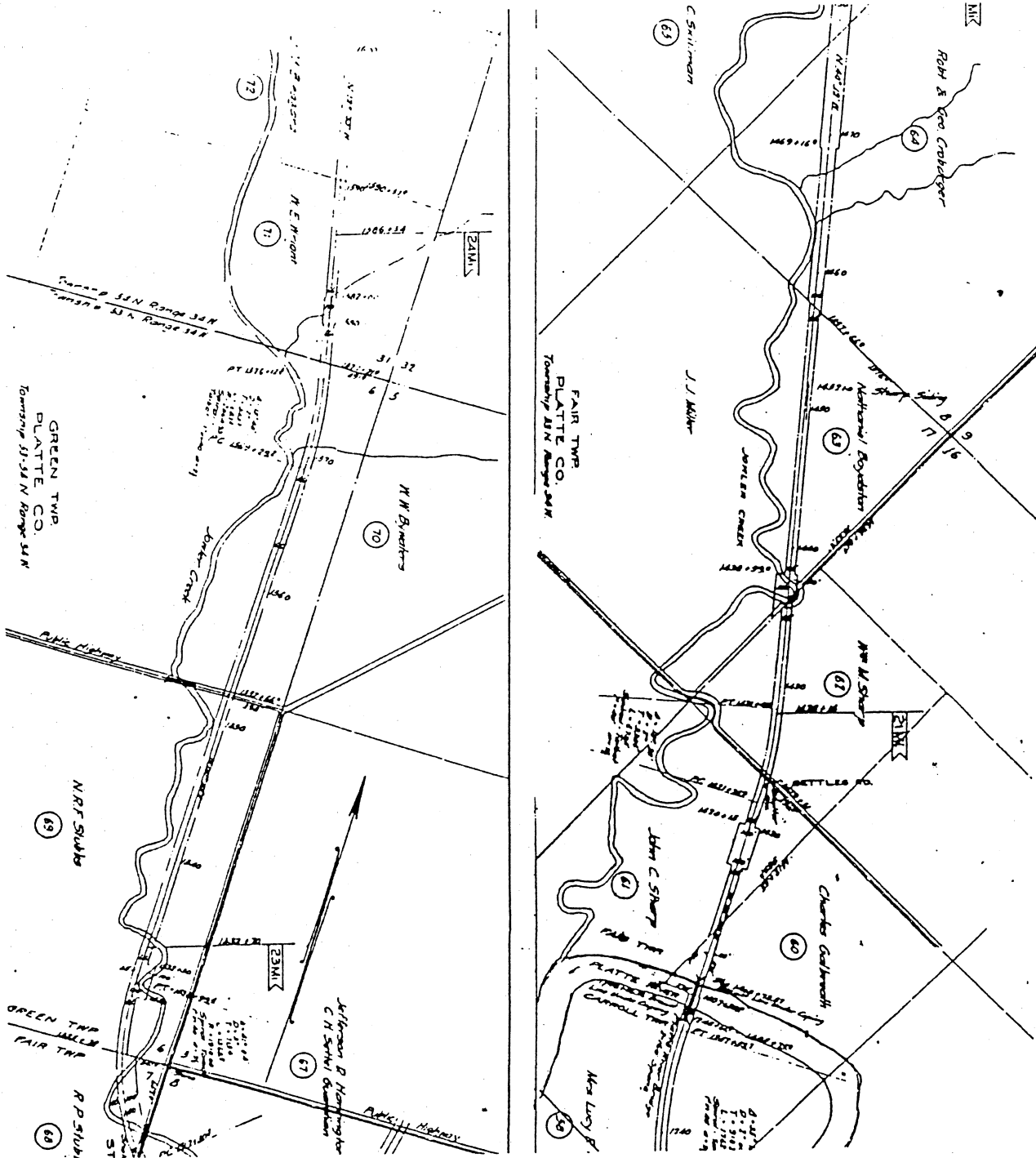
Wyandotte Construction Company Index to Profile and Alignment map, June 1913, showing the route of the St. Joseph line of the KCCC&SJ Railway. Part 2 of 3, showing the center portion of the map. The Jowler Creek Bridge location is noted. (Part 1 on preceding page; Part 3 on following page.)



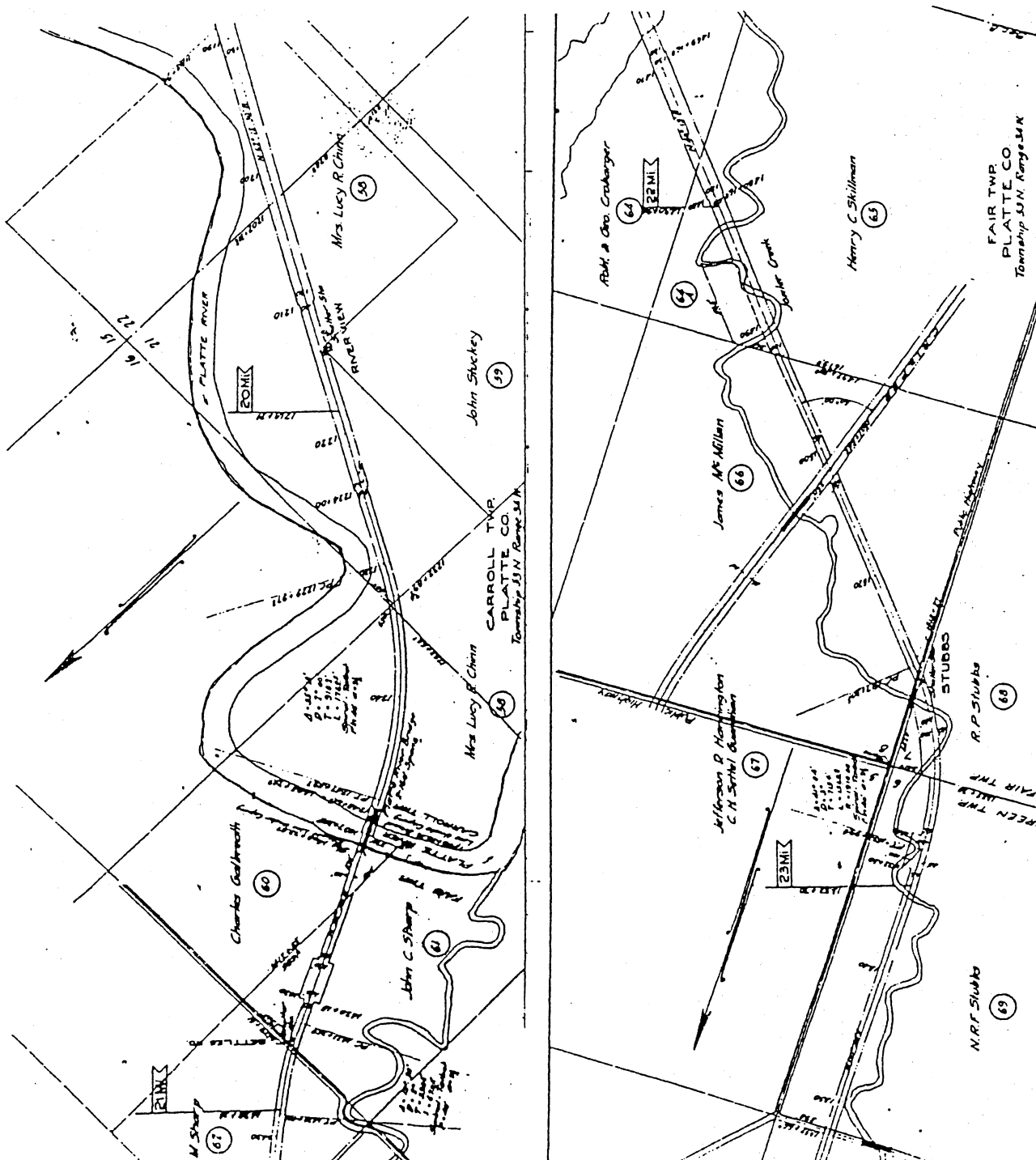
Wyandotte Construction Company Index to Profile & Alignment map, June 1913, showing the route of the St. Joseph line of the KCCC&SJ Railway. Part 3 of 3, showing the lower portion of the map. (Parts 1 and 2 on preceding pages.)



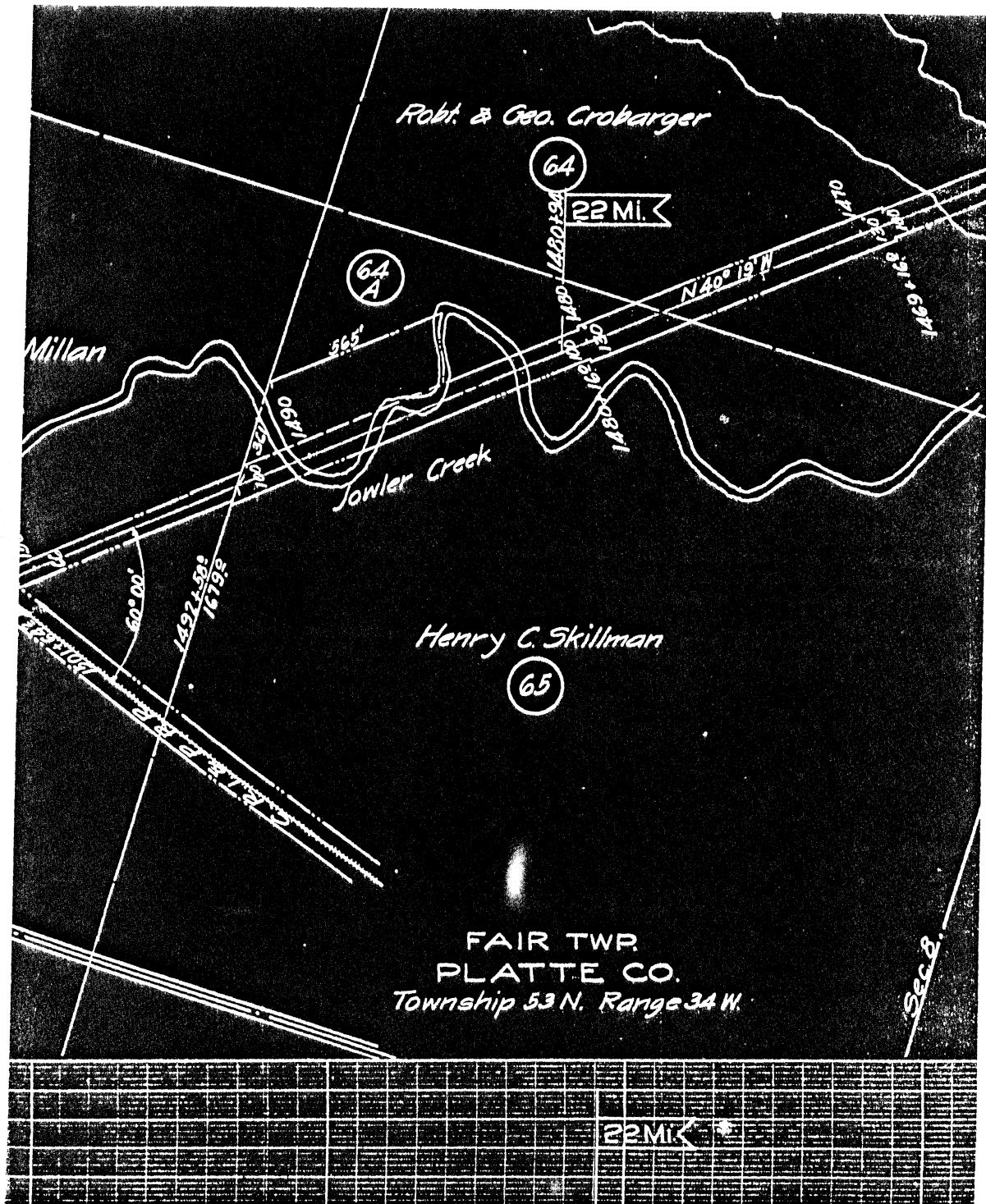
Wyandotte Construction Company Index to Profile & Alignment map, June 1913, Sheet 13, showing the alignment of the St. Joseph line of the KCCC&SJ Railway in the Jowler Creek vicinity. Part 1 of 2, showing the left side of the map. (Part 2 on following page.)



Wyandotte Construction Company Index to Profile & Alignment map, June 1913, Sheet 13, showing the alignment of the St. Joseph line of the KCCC&SJ Railway in the Jowler Creek Bridge vicinity, with bridge location noted. Part 2 of 2, showing the right side of the map. (Part 1 on preceding page.)



Wyandotte Construction Company Profile & Alignment map, June 1913, Sheet 13, showing the profile of the St. Joseph line of the KCCC&SJ Railway. Part 1 of 2, showing the upper portion of the map. (Part 2 on following page.)



Wyandotte Construction Company Profile & Alignment map, June 1913, Sheet 13, showing the profile of the St. Joseph line of the KCCC&SJ Railway. Part 2 of 2, showing the lower portion of the map. (Part 1 on preceding page.)

